

Financial Analysis and Decision Making

Global BBA Fall 2026

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Session 3: Investment Decision Rules

Housekeeping

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In-Class Test

Details

- ▷ Next week, October 1
- ▷ Bring a calculator!
- ▷ Includes material from today, but no Excel
- ▷ Target ~ 45 minutes

Review: NPV Rule

Definition. $NPV = PV \text{ of benefits} - PV \text{ of costs}$

NPV Rule. Invest if $NPV > 0$

- ▷ If projects are mutually exclusive (you can only pick one or the other), choose the project with the higher NPV.
- ▷ If there are no constraints, invest in all projects with a positive NPV.

When can constraints matter?

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-10
1	120	21
NPV	9.09	9.09

When can constraints matter?

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-100
1	120	0
2	0	0
3	0	0
4	0	0
5	0	175.69
NPV	9.09	9.09

Internal Rate of Return

Definition. The **internal rate of return** (IRR) is the discount rate at which the NPV is equal to zero.

Normally, we compute the NPV as

$$\text{NPV} = -I + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \frac{C_3}{(1+r)^3} + \dots$$

To compute the IRR, we set the NPV to zero and solve for the IRR

$$0 = -I + \frac{C_1}{1+\text{IRR}} + \frac{C_2}{(1+\text{IRR})^2} + \frac{C_3}{(1+\text{IRR})^3} + \dots$$

(using Excel)

The IRR Rule

The IRR is often interpreted as the average rate of return on an investment.

Example. Suppose you have an investment project that requires an initial investment of 100 but delivers 115 in one year. Assume $r = 10\%$.

- a. Compute the NPV of this project
- b. Compute the IRR



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The IRR Rule

The IRR Rule. Invest if the project has a IRR greater than the company's cost of capital k .

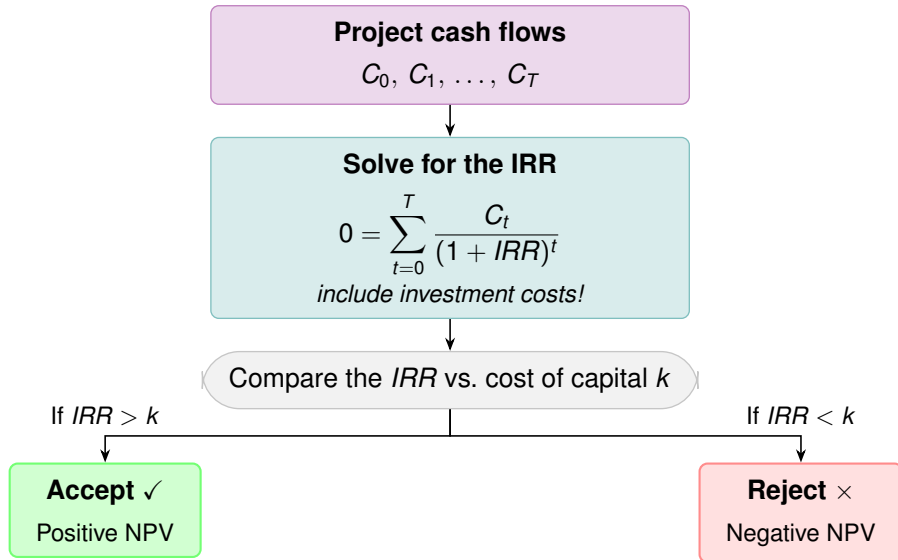
- Intuition: interpret the IRR as the average rate of return on a project. If the company's cost of capital exceeds the IRR, this implies the project's return is below the corporation's cost of financing / required rate of return.



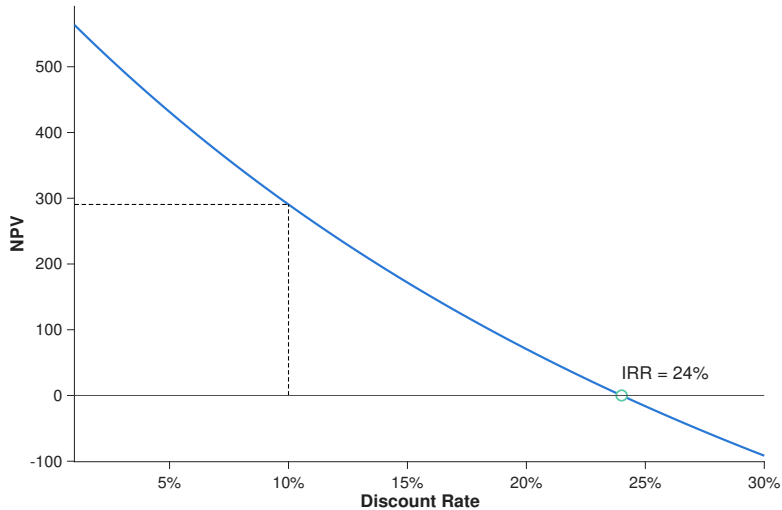
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The IRR Rule



NPV as a function of the IRR



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Pitfalls of IRR: Scale

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-10
1	120	12
NPV	9.09	.91
IRR	20%	20%

Both projects have the same IRR, but project A has a higher NPV. IRR ignores **scale**.

Pitfalls of IRR: Timing

Suppose we have the following two projects and their stream of cash flows. Assume $r = 10\%$.

Year	Project A	Project B
0	-100	-100
1	175	50
2	0	75
3	0	100
NPV	59.1	82.57
IRR	75%	47%

Project B has a higher NPV but Project A has the higher IRR. IRR is sensitive to the **timing** of cash flows.

Pitfalls of IRR: Timing



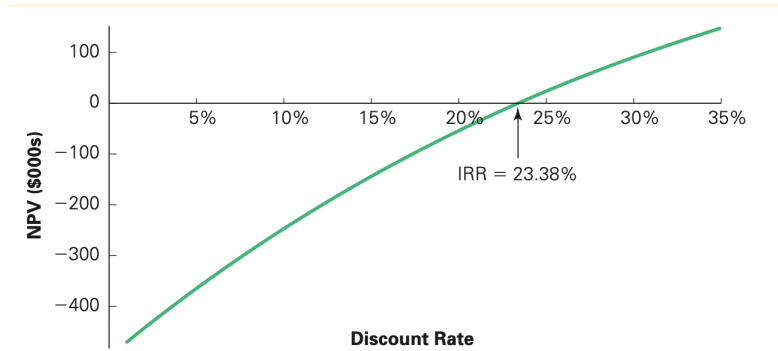
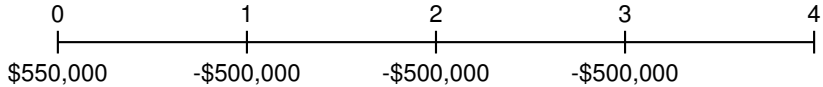
“Our equity strategy managed \$189 billion of AUM as of December 31, 2025 ... generating 39% gross IRR and a 24% net IRR on a compound annual basis.”

Apollo Global Management 2025 10-K

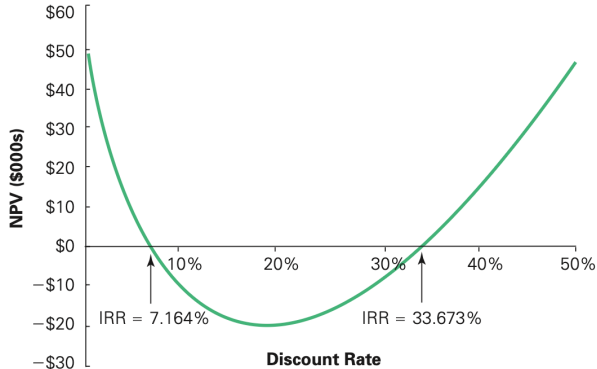
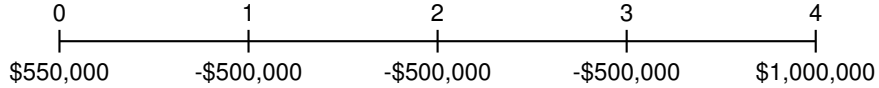
- ▷ That's \$28 bn in performance fees!

The nature of PE can generate high IRR by front-loading cash flows via recapitalizations through debt or cash.

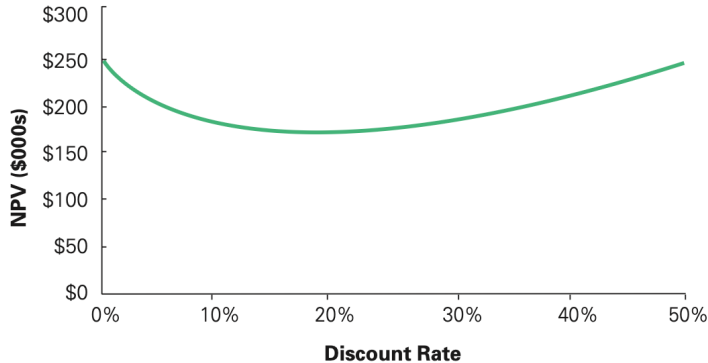
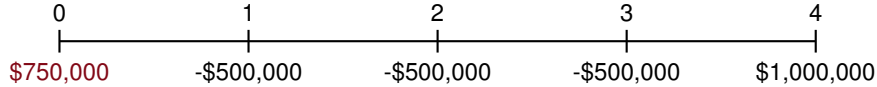
When the IRR breaks down: Delayed Investment



When the IRR breaks down: Multiple IRRs



When the IRR breaks down: Nonexistent IRR



Excel Functions

- ▷ `-PV(rate,nper,pmt)` & `-FV(rate,nper,pmt)`
 - Gives the PV or FV of an annuity
- ▷ `=NPV(rate,value1,[value2],...)`
 - Assumes [value1] is C_1
- ▷ `=PMT(rate,nper,pv)`
 - Solves for C in the annuity formula